

Recall from last class:

- **Theorem:** Let $\mathcal{B} = \{b_1, \dots, b_n\}$ be a basis for a vector space V . Then for each $x \in V$ there exists unique scalars $c_1, \dots, c_n$ such that $x = c_1b_1 + \dots + c_nb_n$.
- Suppose $\mathcal{B} = \{b_1, \dots, b_n\}$ is a basis for a vector space V . The **coordinates of x relative to $\mathcal{B}$** are the weights $c_1, \dots, c_n$ such that $x = c_1b_1 + \dots + c_nb_n$. We define the **coordinate mapping** from V to $\mathbb{R}^n$ by $[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$.
- Let $\mathcal{B} = \{b_1, \dots, b_n\}$ be a basis for $\mathbb{R}^n$, then define $P_{\mathcal{B}} = [b_1 \ \dots \ b_n]$. We call $P_{\mathcal{B}}$ the **change of coordinates matrix**. Note $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$ and $[x]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}x$.
- **Theorem:** Let $\mathcal{B} = \{b_1, \dots, b_n\}$ be a basis for a vector space V . Then the coordinate mapping $x \mapsto [x]_{\mathcal{B}}$ is a one-to-one, onto, linear transformation from V to $\mathbb{R}^n$.

Definition 1. Let $T : V \rightarrow W$ be a linear map of vector spaces. If T is one-to-one and onto, then T is a **vector space isomorphism**, and we say V and W are isomorphic. We write $V \cong W$.

1. Show that $\mathbb{P}_3$ and $\mathbb{R}^4$ are isomorphic.

Theorem: If a vector space V has a basis $\mathcal{B} = \{b_1, \dots, b_n\}$, then any set in V containing more than n vectors is linearly dependent.

2. Prove the theorem.

Cor: If V is a vector space with basis $\mathcal{B} = \{b_1, \dots, b_n\}$, then each set of linearly independent vectors has at most n elements.

Theorem: If a vector space V has a basis with n elements, then every basis of V must have exactly n vectors.

3. Prove the theorem.

Definition 2. If V is spanned by a finite set, then V is called **finite dimensional**, and the **dimension** of V , written $\dim(V)$, is the number of vectors in a basis of V . Define $\dim(\{0_V\}) = 0$. If V is not spanned by any finite set, then V is called **infinite dimensional**.

4. Find the dimensions of the following vector spaces.

(a) $\mathbb{R}^n$

(b) $\mathbb{P}_n$

(c) Let $\mathbb{P}$ be the vector space of polynomials.

(d) $H = \left\{ \begin{bmatrix} s - 2t \\ s + t \\ 3t \end{bmatrix} : s, t \in \mathbb{R} \right\}.$

(e) $H = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}.$

(f) $H = \{A \in M_{n \times n} : A^T = A\}.$

Theorem: Let H be a subspace of a finite dimensional vector space V . Any linearly independent set in H can be expanded to a basis for H , and H is finite dimensional and $\dim(H) \leq \dim(V)$.

5. Prove the theorem.

Basis Theorem: Let V be an n -dimensional vector space with $n \geq 1$. Any linearly independent set of exactly n vectors in V is a basis for V . Any set of exactly n vectors that spans V is a basis for V .

6. Prove the theorem.

Definition 3. Let A be an $m \times n$ matrix with rows $a_1, \dots, a_m$. Define the **row space** of A by

$$\text{Row}(A) = \text{Span}\{a_1, \dots, a_m\}.$$

Theorem: If A is row equivalent to B , then $\text{Row}(A) = \text{Row}(B)$. Furthermore, if B is in reduced row echelon form, the non-zero rows of B form a basis for $\text{Row}(A)$ and $\text{Row}(B)$.

7. Prove the theorem.

8. Find the dimensions of $N(A)$, $\text{Col}(A)$, and $\text{Row}(A)$ where

$$A = \begin{bmatrix} 1 & -6 & 9 & 0 & -2 & 0 \\ 0 & 1 & 2 & -4 & 5 & 0 \\ 0 & 0 & 0 & 5 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Definition 4. The **rank** of the $m \times n$ matrix A is the dimension of $\text{Col}(A)$. We write $\text{rank}(A) = \dim(\text{Col}(A))$. The **nullity** of A is the dimension of $N(A)$.

Rank-Nullity Theorem: Let A be an $m \times n$ matrix.

(a)

$$\dim \text{Row}(A) = \dim \text{Col}(A)$$

(b)

$$\text{rank}(A) + \dim N(A) = n$$

9. Prove the theorem

10.

(a) Let A be a 7 by 5 matrix. What is the largest possible rank?

(b) Let A be a 3 by 7 matrix. What is the smallest possible nullity?

(c) Let A be a 5 by 6 matrix. Suppose the solutions to $Ax = 0$ are all multiple of one non-zero vector. Will the matrix equation $Ax = b$ always be consistent?

Invertible Matrix Theorem Continued The following are equivalent to A being invertible:

- m. The columns of A form a basis of $\mathbb{R}^n$.
- n. $\text{Col}(A) = \mathbb{R}^n$.
- o. $\dim \text{Col}(A) = n$.
- p. $\text{rank}(A) = n$.
- q. $N(A) = \{0\}$.
- r. $\dim N(A) = 0$.

11. Prove the theorem.

Theorem: Let $T : V \rightarrow W$ be a linear transformation between vector spaces. If V is finite dimensional, then

$$\dim \text{im}(T) + \dim \ker(T) = \dim(V).$$